

vices made feasible by the Internet. Peer-to-peer computing that enables unrelated users to dedicate portions of their computers to cooperative processing via the Internet is a related phenomenon used mostly by consumers and businesses. This harnesses a potentially large quantity of computing power in the form of excess, spare or dedicated system resources from the complete range of computers spread out across the Internet.

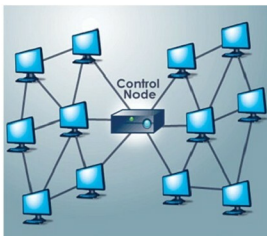
Grid-related technologies can change the way organisations deal with multifaceted computational problems. Many grids are constructed by using clusters or traditional parallel systems as their nodes. For example, World Wide Grid, used in evaluating Gridbus technologies and applications, has many nodes that are clusters.

Cluster computing is made up of multiple interconnected and independent nodes that cooperatively work together as a single unified resource and, unlike grids, cluster resources are owned by a single organisation, and are managed by a centralised resource management and scheduling system that manages allocation of resources to application jobs.

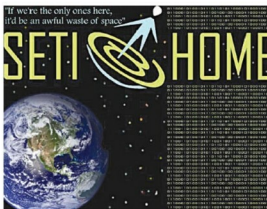
Organisations that are computational power dependent to further their business objectives often have to drop or scale back new projects, design ideas or innovations, simply due to lack of computational bandwidth. Frequently, project demand exceeds computational power supply, despite considerable investments in dedicated computing resources. Upgrading and purchasing new hardware is an expensive option and could soon run into obsolescence, given the rapidly-changing technology.

Grid computing is a better solution given the better utilisation

and distribution of available IT resources. It can be used in executing science, engineering, industrial and commercial applications such as data mining, financial modelling, drug designing, automobile designing, crash simulation, aerospace



In a basic grid computing system, every computer can access the resources of every other computer belonging to the network (Image courtesy: <http://computer.howstuffworks.com>)



SETI@home project analyses telescope data for the search for extraterrestrial intelligence (SETI) (Image courtesy: www.spintrackshop.com)

modelling, high-energy physics, astrophysics, Earth modelling and so on.

Applications of grid computing

SETI@home project. University of California at Berkeley, USA, coordinates one popular scientific example of grid computing, which is an Internet community application that uses background or down-time resources from thousands

of systems, many of these home desktops, to analyse telescope data for the search for extraterrestrial intelligence (SETI) project.

The project's focus is to search for radio signal fluctuations that may indicate a sign of intelligent life from Space. SETI@home is one of the largest and most successful of all Internet-distributed computing projects. Launched in May 1999 to sift through signals collated by Arecibo Radio Telescope in Puerto Rico (the world's biggest radio telescope), the project originally received far more terabytes of data on a daily basis than its assigned computers could handle. Therefore the project directors looked up to volunteers, inviting individuals to download SETI@home software to spare idle processing time on their computers for the project.

After posting a backlog of data, SETI@home volunteers began processing present segments of radio signals captured by the telescope. Currently, about 40GB of data is downloaded every day by the telescope and transmitted to computers all over the globe, to be analysed. Results are sent back via the Internet, and the program then collects a fresh segment of radio signals for the PC to process.

More than two million people—the largest number of volunteers for any Internet-distributed computing project—till date have installed SETI@home software. This global network of three million computers averages about 14 TeraFLOPS, or 14 trillion floating point operations per second, and has gathered more than 500,000 years of processing time in the last two years. Using traditional methods to achieve similar throughput would have